

Cloud Computing

Unit I — Complete Study Notes

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Syllabus Coverage

Importance of Cloud Computing, Characteristics, Pros & Cons, Migrating into the Cloud, Seven-Step Migration Model, Trends in Computing, Cloud Service Models (SaaS, PaaS, IaaS), Cloud Architecture (Logical, Reference Model, System Architecture), Cloud Deployment Models

1. History & Introduction of Cloud Computing

1.1 Historical Evolution

The evolution of cloud computing followed a natural progression from centralized to distributed computing paradigms:

- Client/Server Computing (Pre-Cloud): All software, data, and controls resided on a central server. Users connected to the server to access resources or run programs.
- Distributed Computing: Computers were networked together and shared resources as needed — a stepping stone toward cloud.
- 1961 — John McCarthy (MIT): Proposed that computing could be sold as a utility, similar to water or electricity. This vision was ahead of its time.
- 1999 — Salesforce.com: Pioneered delivering enterprise applications over the Internet via a simple website — making utility computing a reality.
- 2002 — Amazon Web Services (AWS): Began offering services like storage, computation, and human intelligence. The Elastic Compute Cloud (EC2) launched in 2006 as the first truly commercial cloud service.
- 2009 — Google Apps: Started providing cloud computing enterprise applications.
- 2009 — Microsoft Windows Azure: Launched, followed by Oracle, HP, and others — marking cloud computing as mainstream.

1.2 Definition of Cloud Computing (NIST)

Cloud Computing

"Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g. networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

1.3 Key Terminologies

IT Resource	<i>An IT-related artifact that may be either physical or virtual. Examples: physical server, virtual server, software program, service, storage device, network device.</i>
On-Premise	<i>When an IT resource is hosted within the organizational boundary of an IT enterprise, it is considered to be located at the premises of that enterprise.</i>
Scaling	<i>The ability of an IT resource to handle continuously increasing or decreasing workloads.</i>
Cloud Service	<i>Any IT resource that can be accessed remotely via a cloud. It is a broad concept — it can be a web-based software application or a virtual server.</i>
Cloud Service Consumer	<i>The external entities outside the cloud boundary that access cloud-based web services.</i>

Horizontal vs. Vertical Scaling

Parameter	Horizontal Scaling	Vertical Scaling
Definition	Adding more machines/nodes to the pool	Adding more CPU/RAM/Disk to existing server
Also Called	Scaling Out / Scaling In	Scaling Up / Scaling Down
Cost	Less expensive	More expensive
Availability	Resources available instantly at all times	Not always instantly available
Additional Setup	Resource replication required	Additional setup normally needed
Hardware Limit	Not limited by hardware capacity	Limited by hardware capacity
Example Users	Google, YouTube, Amazon, Facebook	Mid-range & small companies

2. Importance & Characteristics of Cloud Computing

2.1 Importance of Cloud Computing

Cloud computing is important because it offers flexibility, data recovery, minimal maintenance, easy access, and increased security. The five key areas are:

1. **Flexibility:** Cloud resources scale up or down based on demand. Organizations pay only for what they use, avoiding over-provisioning.
2. **Data Recovery:** Physical server damage is no longer catastrophic. Data on the cloud can be recovered without dependence on a physical device.
3. **No Maintenance:** The cloud provider handles all hardware maintenance, replacements, and upgrades. The organization's only expenditure is the service subscription.
4. **Easy Access:** Employees can work remotely from any device with network access. All documents and backend files are accessible through the cloud interface.
5. **Increased Security:** Data stored on the cloud is not dependent on a single physical device. Even if a device is lost or stolen, data remains accessible and is protected through encryption and login security.

2.2 Significance of Cloud Computing

- Applications are made available to everyone through the cloud as and when needed.
- IT costs are reduced due to fewer servers, lower software costs, and reduced staffing needs.
- Investment costs are more flexible compared to traditional methods.
- Increased reliability — anyone can get connected anytime.
- Data and applications are mobile and easily made available to employees across the globe.
- Increased collaboration reduces time-to-market and accelerates product enhancement.
- Facility can be turned up, down, or off as per demand — elastic provisioning.
- Delivers better integration with Mergers & Acquisitions (M&A) activity.
- Increased 'green' credentials due to lesser environmental impact.

2.3 Characteristics of Cloud Computing

On-Demand Self-Service	<i>A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.</i>
Broad Network Access	<i>Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).</i>
Resource Pooling	<i>The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.</i>

Rapid Elasticity

Resources can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand.

Measured Service

Cloud systems automatically control and optimize resources by leveraging a metering capability at some level of abstraction appropriate to the type of service. It provides an analyzable and predictable computing platform.

2.4 Advantages of Cloud Computing

- **Additional CPU cycles or storage can be acquired on demand without buying physical equipment. Once the need is fulfilled, the service is simply stopped.:** Increased Scalability
- **No need to buy and configure new equipment; proportional, pay-per-use cost model.:** Simplicity / Reduced Investments
- **First adopters include Amazon, Google, Microsoft, IBM, and Yahoo — all reputable companies offering reliable, high-capacity service.:** Knowledgeable Vendors
- **A cloud-based IT resource can be accessible for longer periods (e.g., approximately 22 hours per day), reducing the effect of runtime failures.:** Increased Availability & Reliability
- **Reputable vendors implement comprehensive encryption and login security to protect data.:** Security

2.5 Disadvantages / Limitations of Cloud Computing

- **Once data leaves your control and is placed with a service provider, a layer of control is lost. Law enforcement has greater access to cloud-stored data than organization-server data.:** Sensitive Information Risk
- **Some applications require excessive bandwidth, are difficult to integrate with other systems, or cannot communicate securely across the Internet.:** Applications Not Ready
- **Hackers target cloud data since it resides on someone else's equipment. Botnets can perform Distributed Denial of Service (DDoS) attacks leading to potential blackmail.:** Security Concerns (Hackers & Bots)
- **Organizations may become dependent on a single cloud vendor, making migration difficult.:** Vendor Dependence

3. Migrating into the Cloud

3.1 Definition of Cloud Migration

Cloud Migration

Cloud migration is the process of moving digital business operations into the cloud. It involves transferring data, applications, and IT processes from on-premises or legacy data centers to cloud infrastructure provided by providers such as AWS, Google Cloud, or Azure.

3.2 Benefits of Migrating to the Cloud

1. **Scalability:** Cloud computing scales up to support larger workloads and more users far more easily than on-premises infrastructure, which requires physical server purchases.
2. **Cost Reduction:** Companies reduce IT operational spending because cloud providers handle maintenance and upgrades. Focus shifts to core business activities.
3. **Performance:** Hosting applications in cloud data centers (closer to users) reduces latency and improves the overall user experience.
4. **Flexibility:** Users — whether employees or customers — can access cloud services and data from anywhere, enabling international reach and flexible working.

3.3 Challenges of Cloud Migration

1. **Migrating Large Databases:** Databases often need to move to a different platform entirely. Some providers offer physical data transfer methods (e.g., shipping hardware appliances) for massive databases.
2. **Data Integrity:** After transfer, ensuring data is intact, secure, and not leaked during the migration process is critical.
3. **Continued Operation:** The business must keep current systems operational and available throughout migration. A copy of all data must exist in the cloud before any existing database is shut down. Migration must happen incrementally.

3.4 Cloud Migration Strategies (5 R's)

Strategy	Description	Target Service Model
Rehost	Move application architecture as-is to cloud servers ("lift and shift")	IaaS
Refactor	Reuse existing code and frameworks but run on a PaaS provider's platform	PaaS
Revise	Partially rewrite or expand the code base, then deploy via rehosting or refactoring	IaaS / PaaS
Rebuild	Rewrite and re-architect the entire application from scratch on PaaS	PaaS
Replace	Discard old applications and switch to ready-built SaaS applications from vendors	SaaS

3.5 Seven-Step Model of Migration into Cloud

Exam Tip

The 7-step model is frequently asked. Remember the sequence: Assessment → Isolating → Mapping → Re-Architect → Augment → Validate → Optimization

1. **Assessment:** Evaluate the cost of migration and the Return on Investment (ROI) achievable for the production version. This is the first and iterative step.
2. **Isolating:** Identify and isolate all systemic and environmental dependencies of enterprise application components within the captive data center. This reveals the complexity level of the migration.
3. **Mapping:** Generate mapping constructs between what will remain in the local captive data center and what will move to the cloud.
4. **Re-Architect:** Redesign and re-implement substantial parts of the enterprise application on the cloud. Note that some functionality may be lost during this process.
5. **Augment:** Leverage internal features of the cloud computing service to augment the enterprise application in small but meaningful ways.
6. **Validate:** Test the new form of the enterprise application with an extensive test suite — covering components both on-cloud and on-premises. Results could be positive or mixed.
7. **Optimization:** Iterate and optimize based on test results. After several optimizing iterations, the migration is deemed successful.

4. Cloud Service Models (SaaS, PaaS, IaaS)

4.1 Software as a Service (SaaS)

SaaS

SaaS (also known as "On-Demand Software") is a software distribution model in which services are hosted by a cloud service provider and are available to end-users over the Internet. End-users do not need to install any software on their devices to access these services.

Types of SaaS Services

- Business Services: ERP (Enterprise Resource Planning), CRM (Customer Relationship Management), billing, and sales.
- Document Management: Software offered by a third-party SaaS provider to create, manage, and track electronic documents. Example: Slack, Box, Zoho Forms.
- Social Networks: Social networking service providers use SaaS to handle the general public's information at scale.
- Mail Services: Email providers (Hotmail, Gmail, Yahoo Mail) deliver services via SaaS to handle unpredictable user loads.

SaaS Categories

1. Line of Business Services: Business solutions offered to companies and enterprises via a subscription. Includes supply-chain management, CRM, and similar tools.
2. Customer-Oriented Services: Offered to the general public on a subscription basis or for free (supported by advertising). Examples: web mail, online gaming, consumer banking.

Characteristics of SaaS

- Makes various types of software available over the Internet.
- The vendor is responsible for maintaining the software.
- Can scale up or down on demand.
- Available anytime, anywhere, on demand.

Advantages of SaaS

- Modest Software Tools: No installation required at the client side.
- Efficient Use of Software Licenses: No need to purchase individual software licenses.
- Centralized Management & Data: Data is saved centrally by the cloud provider.
- Platform Responsibilities Managed by Provider: All platform responsibilities are handled by the cloud provider.
- Multi-Tenant Solutions: Enable multiple users to share a single instance of multiple resources in virtual isolation.

Disadvantages of SaaS

- Browser-Based Risks: Security vulnerabilities through web browsers.
- Network Dependence: Requires a reliable Internet connection.

- Lack of Portability: Difficult to migrate between different SaaS cloud providers.

4.2 Platform as a Service (PaaS)

PaaS

Platform as a Service (PaaS) provides a runtime environment. It allows programmers to easily create, test, run, and deploy web applications. It is purchased on a pay-as-per-use basis and accessed via the Internet. Back-end scalability is managed by the cloud service provider.

PaaS includes infrastructure (servers, storage, and networking) AND platform (middleware, development tools, database management systems, business intelligence) to support the web application life cycle.

Examples: Google App Engine, Force.com, Joyent, Microsoft Azure.

Key Features of PaaS

- Provides a ready-made runtime environment for different types of applications.
- Offers development and deployment tools necessary to build applications.
- Provides point-and-click tools useful for non-developers to create web applications.
- Developers can log on and use built-in APIs for web-based application creation.

Advantages of PaaS

- Low Overhead of Administration: Administration is the cloud provider's responsibility.
- Less Ownership Cost: No need to purchase expensive hardware, servers, or data storage.
- Scalable Solutions: Resources can be scaled up or down automatically and flexibly.
- Latest System Software: The cloud provider maintains and keeps advanced, latest software versions.

Disadvantages of PaaS

- Lack of Portability: Difficult to migrate between different PaaS clouds.
- Resource Constraints: Event-based processor scheduling imposes time limitations on answering a request.
- Dependency on Network: PaaS applications are completely reliant on the network and must explicitly manage cryptography and security exposures.

4.3 Infrastructure as a Service (IaaS)

IaaS

IaaS (also known as Hardware as a Service — HaaS) is a cloud computing layer that allows customers to outsource their IT infrastructures such as servers, networking, virtual machines, processing, and storage resources. Customers access these resources via the Internet using a pay-as-per-use model.

Core Components of IaaS

- Compute: Virtual Central Processing Units (vCPUs) and virtual main memory for VMs provisioned to end-users.
- Storage: Back-end storage for storing files provided by the IaaS provider.
- Network (NaaS): Networking components such as routers, switches, and bridges for VMs.
- Load Balancers: Provides load balancing capability at the infrastructure layer.

Additional IaaS Facilities

- Virtual machine disk storage
- Software bundles
- Virtual Local Area Networks (VLANs)
- IP addresses
- Load balancers

Server Types in an IaaS Solution

- Physical Server: Actual hardware allocated for the dedicated use of a customer.
- Dedicated Virtual Server: A virtual server allocated to a customer that runs on a physical server (which may or may not host other virtual servers).
- Shared Virtual Server: A virtual server on a device shared with multiple customers.

Advantages of IaaS

- Allows IT to Shift Focus: Fast accessibility to compute resources via a utility/subscription model.
- Dynamic Scaling: Infrastructure services can be scaled up and down; supports multiple tenants.
- Investment Cap: Highly beneficial for businesses with limited capital to invest in hardware.
- Flexible Offering: Access infrastructure from any location, any device.

Disadvantages of IaaS

- Security: Most IaaS providers cannot provide 100% security guarantee.
- Maintenance & Upgrade: IaaS providers maintain but do not always upgrade software for all organizations.
- Interoperability Issues: Difficult to migrate VMs from one IaaS provider to another — potential vendor lock-in.

4.4 IaaS vs. PaaS vs. SaaS — Comparison Table

Parameter	IaaS	PaaS	SaaS
Provides	Virtual data center for storage, networking, compute	Virtual platforms and tools to create, test, deploy apps	Web software and apps to complete business tasks
Gives Access To	Virtual machines, virtual storage, IP, load balancers	Runtime environments and deployment tools	Software as a service to end-users
Used By	Network Architects	Developers	End-Users

Stack Provided	Infrastructure only	Infrastructure + Platform	Infrastructure + Platform + Software
Examples	AWS EC2, Microsoft Azure VM	Google App Engine, Force.com	Gmail, Salesforce, Slack

5. Cloud Computing Architecture

5.1 Definition

Cloud Computing Architecture

Cloud computing architecture is a combination of service-oriented architecture (SOA) and event-driven architecture. It is divided into two parts: Front End (client-side) and Back End (service provider side), connected via the Internet.

5.2 Front End

The front end is used by the client. It contains client-side interfaces and applications that are required to access the cloud computing platforms. Components include:

- Web Browsers (Chrome, Firefox, Internet Explorer, etc.)
- Thin Clients (computers without internal hard drives that use the server for processing)
- Fat/Thick Clients (regular computers with full capabilities using a browser to access the cloud)
- Tablets and Mobile Devices

5.3 Back End Components

Application	<i>Refers to the software or platform to which the client accesses. It provides services as per client requirements.</i>
Service	<i>Refers to the three major cloud-based service types — SaaS, PaaS, and IaaS — and manages which type of service the user accesses.</i>
Cloud Runtime	<i>Provides the execution and runtime platform/environment to the virtual machine.</i>
Storage	<i>Provides flexible and scalable storage service and management of stored data.</i>
Infrastructure	<i>Hardware and software components of cloud such as servers, storage, network devices, and virtualization software.</i>
Management	<i>Manages back-end components like application, service, runtime cloud, storage, infrastructure, and security mechanisms.</i>
Security	<i>Implements different security mechanisms in the backend for secure cloud resources, systems, files, and infrastructure to end-users.</i>
Internet	<i>Acts as the medium (bridge) between front end and back end, establishing interaction and communication between both.</i>

5.4 Components of Cloud Computing Architecture

1. Hypervisor

- A virtual machine monitor that provides virtual operating platforms to every user.
- Manages guest operating systems in the cloud.

- Runs a separate virtual machine on the backend consisting of software and hardware.
- Main objective: divide and allocate resources among virtual machines.

2. Management Software

Responsible for managing and monitoring cloud operations with various strategies to increase cloud performance. Key operations include:

- Compliance auditing
- Management of disaster oversight
- Contingency planning

3. Deployment Software

- Contains all mandatory installations and configurations required to run a cloud service.
- Every cloud service deployment is performed using deployment software.
- Three deployment models: SaaS (e.g., Gmail), PaaS (e.g., Microsoft Azure), IaaS (pay-as-you-go pricing).

4. Network

- Connects the front-end and back-end and allows every user to access cloud resources.
- Helps users connect and customize the route and protocol.
- A virtual server hosted on the cloud computing platform. Highly flexible, secure, and cost-effective.

5. Cloud Storage

- Every bit of data is stored and accessed by a user from anywhere over the Internet.
- Scalable at runtime and automatically accessed.
- Data can be modified and retrieved from cloud storage over the web.

5.5 Benefits of Cloud Computing Architecture

- Makes the overall cloud computing system simpler.
- Improves data processing requirements.
- Helps in providing high security.
- Makes it more modularized.
- Results in better disaster recovery.
- Gives good user accessibility.
- Reduces IT operating costs.

6. Cloud Computing Reference Model

6.1 Definition

Cloud Computing Reference Model

The reference model for cloud computing is an abstract model that characterizes and standardizes a cloud computing environment by partitioning it into abstraction layers and cross-layer functions.

6.2 Layers in the Reference Model

1. Infrastructure as a Service (IaaS): The foundational layer providing access to computing resources such as servers, storage, and networking.
2. Platform as a Service (PaaS): Built on top of IaaS; provides a cloud environment to develop, manage, and deliver applications.
3. Software as a Service (SaaS): Built on top of PaaS; provides access to vendor's cloud-based software. Applications reside on a remote cloud network accessed through the web or an API.
4. Web 2.0: A layer that allows anyone to create and share online information. A key element is that it allows people to create, share, collaborate, and communicate.

6.3 Virtualization and Cloud Computing

Virtualization

Virtualization is the partitioning of a single physical server into multiple logical servers. Each logical server behaves like a physical server and can run an operating system and applications independently. It is the main enabling technology for cloud computing.

Virtualization is used for three main purposes:

1. Network Virtualization: A method of combining available resources in a network by splitting bandwidth into independent channels. Each channel can be assigned to a specific server or device in real time.
2. Storage Virtualization: Pooling physical storage from multiple network storage devices into what appears to be a single storage device, managed from a central console. Commonly used in Storage Area Networks (SANs).
3. Server Virtualization: The masking of server resources like processors, RAM, and OS from server users, intending to increase resource sharing and reduce computation burden.

7. Cloud Deployment Models

7.1 Overview

Cloud deployment models are categorized based on their location and who has access to the infrastructure. Most cloud hubs have tens of thousands of servers and storage devices to enable fast loading.

7.2 Public Cloud

Public Cloud

Public cloud is open to all to store and access information via the Internet using the pay-per-usage method. Computing resources are managed and operated by the Cloud Service Provider (CSP). Examples: Amazon EC2, IBM SmartCloud Enterprise, Microsoft Azure, Google App Engine, Windows Azure Services Platform.

Advantages

- Lower cost than private and hybrid cloud.
- Maintained by the CSP — no maintenance burden on the user.
- Easier to integrate — offers better flexibility to consumers.
- Location independent — services delivered through the Internet.
- Highly scalable as per computing resource requirements.
- No limit on number of users — accessible by the general public.

Disadvantages

- Less secure — resources are shared publicly.
- Performance depends on high-speed internet network link to the cloud provider.
- The client has no control over the data.

7.3 Private Cloud

Private Cloud

Private cloud (also known as internal cloud or corporate cloud) is used by organizations to build and manage their own data centers internally or by a third party. It can be deployed using open-source tools such as OpenStack and Eucalyptus. NIST divides it into: (1) On-premise private cloud and (2) Outsourced private cloud.

Advantages

- High level of security and privacy.
- Better performance with improved speed and space capacity.
- IT team can quickly allocate and deliver on-demand IT resources.
- Organization has full control — no dependence on external parties.
- Suitable for organizations where data security is the first priority.

Disadvantages

- Requires skilled people to manage and operate cloud services.
- Limited area of operations — only accessible within the organization.
- Not suitable for organizations with a high user base or insufficient pre-built infrastructure.

7.4 Hybrid Cloud

Hybrid Cloud

Hybrid Cloud = Public Cloud + Private Cloud. It is partially secure: services running on the public cloud can be accessed by anyone, while services on the private cloud can be accessed only by the organization's users. Examples: Google Application Suite, Office 365, Amazon Web Services.

Advantages

- More security than pure public cloud.
- Helps deliver new products and services more quickly.
- Excellent way to reduce risk.
- Flexible resources (from public cloud) + Secure resources (from private cloud).

Disadvantages

- Security features are not as good as private cloud.
- Managing a hybrid cloud is complex — managing more than one deployment model is difficult.
- Reliability depends on cloud service providers.

7.5 Community Cloud

Community Cloud

Community cloud allows systems and services to be accessible by a group of several organizations to share information between organizations and a specific community. It is owned, managed, and operated by one or more organizations in the community, a third party, or a combination. Example: Healthcare Community Cloud.

Advantages

- Cost-effective — the cloud is shared by several organizations or communities.
- Suitable for organizations wanting collaborative cloud with more security than public cloud.
- Better security than public cloud.
- Provides collaborative and distributive environment.
- Allows sharing of cloud resources, infrastructure, and capabilities among various organizations.

Disadvantages

- Not a good choice for every organization.
- Security features are not as good as the private cloud.
- Not suitable if there is no collaboration between community members.
- Fixed amount of data storage and bandwidth is shared among all community members.

7.6 Deployment Model Comparison

Parameter	Public Cloud	Private Cloud	Hybrid Cloud	Community Cloud
Host	Service Provider	Enterprise / Third Party	Enterprise / Third Party	Community / Third Party
Users	General Public	Selected Users	Selected Users	Community Members
Access	Internet	Internet, VPN	Internet, VPN	Internet, VPN
Owner	Service Provider	Enterprise	Enterprise	Community
Cost	Lowest	Highest	Medium	Shared / Low
Security	Lowest	Highest	Medium-High	Medium

8. Cloud Computing Components & Supporting Concepts

8.1 Cloud Components

A cloud computing solution is made up of three core elements: Clients, Datacenter, and Distributed Servers.

Clients	<i>Devices that end-users interact with to manage information on the cloud. Three types: Mobile (PDAs, smartphones), Thin Clients (no hard drives; server does all processing), Thick Clients (regular computers using a web browser to connect to the cloud).</i>
Datacenter	<i>The collection of servers where the subscribed application is housed. It could be a large room in the same building or servers across the world accessed via the Internet.</i>
Distributed Servers	<i>Servers that are not housed in the same location but are distributed geographically for redundancy and performance.</i>

8.2 Load Balancing

Load balancing is a technique used by high-traffic sites (Google, Yahoo, Amazon) to manage huge web requests by sharing them across multiple servers.

- A load balancing server distributes web requests to two or more servers.
- For complex server applications involving databases, shared database access and replicas are required.
- Database software synchronizes data updates across all servers to maintain data consistency.
- Cloud-based database or cloud-based Network Attached Storage (NAS) simplifies this — the cloud handles synchronization and replication internally.

8.3 System and Storage Redundancy

- Cloud-based computing does not constrain physical resource locations.
- Companies use collocation — employing duplicate off-site servers or disk storage devices.
- Duplicate resources allow control transfer from one location to another or use for load balancing.
- Benefits: reduced susceptibility to fire, natural disasters, terrorism, downtime from power loss; improved performance through workload distribution.
- IaaS providers allow organizations to automatically scale resource allocation up or down.

8.4 Network Attached Storage (NAS)

NAS	<i>A NAS unit is a computer connected to a network that provides only file-based data storage to other devices on the network. It is not designed to be a general-purpose server. NAS systems contain one or more hard disk drives, often arranged into logical, redundant storage containers (RAID). Users can scale cloud-based storage on demand and pay only for consumed storage.</i>
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8.5 Identity as a Service (IDaaS)

IDaaS

Identity as a Service (IDaaS) is a cloud-based service that provides identity and access management (IAM) functions to target systems on a customer's premises and/or in the cloud. It is a cloud-oriented third-party authentication service offered on a subscription-based managed service model.

IDaaS is used for electronic transactions, enabling remote erasure of employee access when they leave the organization. It addresses the tedious, time-consuming, and expensive nature of traditional ID management.

Single Sign-On (SSO)

Single Sign-On (SSO)

SSO is an authentication process with which a user can access multiple applications using only one set of login information. Through SSO, a user does not have to log in repeatedly to different systems.

SSO Process (5 Steps):

1. User logs into the authentication server by entering a valid username and password.
2. The authentication server returns a user ticket.
3. The user sends the ticket to the intranet server.
4. The intranet server sends the ticket to the authentication server for verification.
5. The authentication server sends the user's security credentials back to the intranet server, granting access.

Note: When an employee leaves the organization, IT staff must disable the user at the authentication server — this immediately disables access to all connected systems.

Advantages of SSO

- Manages local and remote applications along with the desktop flow.
- Removes re-authentication and improves productivity.
- Flexible and user-friendly database management.
- Serves detailed user access reporting.

8.6 Service-Oriented Architecture (SOA)

SOA

A service-oriented architecture is a group of services that interact with each other. The interaction can involve either data transfer or synchronization of one or more services into a particular action.

Service

A service is defined as a function that is definite and independent of the framework or condition of other services.

Three Building Blocks of SOA:

- Service Provider: The owner of the service; the platform that hosts access to the service.
- Service Requestor: The business that requires certain functions to be satisfied; the application that looks for and invokes an interaction with a service.
- Service Registry: A searchable registry of service descriptions where service providers publish their service descriptions.

Operations in SOA:

- Publish: Service descriptions need to be published so that service requestors can find them.
- Find/Discovery: Service requestor retrieves a service description directly or queries the service registry for the required service.
- Bind/Access: Service requestor invokes or initiates an interaction with the service at runtime.

8.7 Multi-Tenancy in SaaS

Multi-Tenancy

A multi-tenant application shares a single stack of resources — both hardware and software — to cater to multiple tenants (organizations, sub-organizations, sections, etc.), satisfying the needs of multiple sub-organizations using a single shared state of software and hardware resources.

A traditional static application is not capable of addressing the unique challenges of multi-tenancy. A multi-tenant application must be dynamic (polymorphic) in nature to fulfill specific expectations of various tenants. Application components are generated at runtime from metadata — data about the application itself. The architecture requires separate components for: Runtime application data, Metadata describing base functionality, and Tenant-specific customization metadata.

9. Quick Revision Glossary

Exam Strategy

All definitions in this glossary carry keyword marks. Write definitions exactly as given during exams. Follow definitions with elaboration in your own words for full marks.

Cloud Computing (NIST)	<i>A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction.</i>
SaaS	<i>On-Demand Software — services hosted by a cloud service provider, available to end-users over the Internet without any installation requirement.</i>
PaaS	<i>Provides a runtime environment allowing programmers to create, test, run, and deploy web applications. Back-end scalability is managed by the cloud service provider.</i>
IaaS (HaaS)	<i>Allows customers to outsource IT infrastructures such as servers, networking, virtual machines, processing, and storage via a pay-as-per-use Internet-based model.</i>
Virtualization	<i>Partitioning of a single physical server into multiple logical servers, each capable of running an OS and applications independently. The main enabling technology for cloud computing.</i>
Multi-Tenancy	<i>A single shared instance of software and hardware resources serving multiple tenants (organizations) in virtual isolation.</i>
Cloud Migration	<i>The process of moving digital business operations — data, applications, IT processes — from on-premises or legacy infrastructure to cloud infrastructure.</i>
Hypervisor (VMM)	<i>A virtual machine monitor that provides virtual operating platforms to users and manages guest operating systems in the cloud.</i>
Public Cloud	<i>Open to all; computing resources managed by CSP; accessed via Internet on pay-per-use basis.</i>
Private Cloud	<i>Internal or corporate cloud used exclusively by an organization; high security and control; deployed via OpenStack or Eucalyptus.</i>
Hybrid Cloud	<i>Combination of public and private cloud (Public + Private). Partially secure.</i>
Community Cloud	<i>Shared by a group of organizations with common concerns; owned by one or more community members, a third party, or a combination.</i>
SSO	<i>Single Sign-On — an authentication process allowing a user to access multiple applications using a single set of login credentials.</i>
IDaaS	<i>Identity as a Service — cloud-based identity and access management offered as a subscription-based third-party authentication service.</i>

Load Balancing

A technique used to distribute web requests across multiple servers to manage high traffic and avoid overload on a single server.

NAS

Network Attached Storage — a computer connected to a network providing only file-based data storage to other devices; usually uses RAID storage and a stripped-down OS.

SOA

Service-Oriented Architecture — a group of services that interact with each other via data transfer or service synchronization.

10. Probable Exam Questions

Note

These are commonly asked questions based on the syllabus and slide content. Answers for all of these are covered in the notes above.

2-Mark / Short Answer Questions

- Define Cloud Computing (NIST definition).
- What is On-Demand Self-Service?
- Differentiate between horizontal and vertical scaling.
- What is a Hypervisor?
- Define Multi-Tenancy.
- What is IDaaS?
- Define SSO. List its advantages.
- What is a Community Cloud?
- What is the difference between Thin Client and Thick Client?

5-Mark / Medium Answer Questions

- Explain any five characteristics of cloud computing.
- Explain the advantages and disadvantages of cloud computing.
- Compare SaaS, PaaS, and IaaS with examples.
- Explain the advantages and disadvantages of Public Cloud.
- Explain the components of Cloud Computing Architecture.
- What is cloud migration? Explain its benefits and challenges.
- Explain the five cloud migration strategies (5 R's).
- Explain any four components of Cloud Computing Architecture.

10-Mark / Long Answer Questions

- Explain the Seven-Step Model of Migration into the Cloud in detail.
- Explain all four Cloud Deployment Models with advantages and disadvantages. Draw the comparison table.
- Explain IaaS in detail with its components, advantages, and disadvantages. Differentiate IaaS from PaaS and SaaS.
- Explain Cloud Computing Architecture with a neat diagram. Describe Front End and Back End components.
- Explain the Cloud Computing Reference Model with all its layers.
- Explain the importance of cloud computing. Write about virtualization in cloud computing.